

The neural signatures of the psychological construct “flow”: A replication study

Martin Ulrich^{a,*}, Filip Niemann^b, Georg Grön^a

^a Section Neuropsychology and Functional Imaging, Department of Psychiatry, Ulm University, Ulm, Germany

^b Cognition, Aging, and Brain Stimulation Lab, Department of Neurology, University Medicine Greifswald, Greifswald, Germany

ARTICLE INFO

Keywords:

Flow experience

fMRI

Replication Bayes factor

Replication

Confirmatory

ABSTRACT

The psychological construct “flow” has received major attention during the last decade by various scientific branches in the fields of psychology and neuroscience. Flow is operationally defined in relation to the boundary conditions of boredom and overload. According to the conditions’ arrangement, two major neural flow effects are of interest, inferred by quadratic trends of neural activation. The inverted U-shaped pattern of neural activation is characterized by greater neural activation during the flow condition relative to boredom and overload, while the U-shaped pattern is the reverse, that is, lower neural activation during flow relative to its boundary conditions. Both effects have repeatedly been reported during the last years and have seen greater scientific resonance, which is why we found it necessary to try to replicate recent findings. A fresh sample of 41 healthy male participants was investigated with BOLD functional magnetic resonance imaging in combination with our flow paradigm. Electrodermal activation served as read-out of the flow effect on the psychophysiological level. Evidence of replication was quantified in terms of the replication Bayes factor. We observed strong replication evidence for electrodermal activation and decisive evidence of replication for both neural flow effects. Aspects of dorsolateral prefrontal cortex, anterior insula and parietal cortex showed inverted U-shaped activation. U-shaped activation was predominant in aspects of medial prefrontal cortex, ventral striatum, amygdala and along the cingulate cortex (subgenual, middle and posterior). Despite its strong replicability, the flow paradigm has been administered in men-only samples so far. Therefore, present results still await empirical replication in women-only samples.

1. Introduction

The subjective experience of flow can arise during physical and mental tasks that fit with the most important prerequisite: an optimal balance between actual task affordances and individual skills and/or abilities. Situations where this prerequisite cannot be met are unlikely to permit the experience of flow, and flanking subjective experiences arise, that is, boredom or overload. Psychological conditions that promote both experiences therefore constitute the boundary conditions against which the emergence of flow is tested because the entire concept of flow is operationally defined relative to these boundary conditions. In cases where individual abilities and/or skills are in balance with changing task affordances and clear goal settings, the resultant experience of flow is marked by the following features (Csikszentmihalyi, 1975, 1990, 2000): high though almost effortless attention, reduced self-referential

processing, sense of control, and the feeling that the activity per se is rewarding (autotelic activity).

During the past years, we have repeatedly reported about the neural correlates and mechanisms of the flow experience (Ulrich et al., 2016a, b; Ulrich et al., 2014; Ulrich et al., 2018). As these studies have received major attention in the meanwhile, the present study was set up with the main intention to test for the replicability of previous imaging results in a fresh sample of 41 healthy participants. As read-out of neural activation we decided to use blood-oxygen-level-dependent (BOLD) functional magnetic resonance imaging (fMRI) instead of MR-based perfusion imaging (as in Ulrich et al., 2014; Ulrich et al., 2018). The former method is of more widespread use than the latter, and the results from perfusion imaging (Ulrich et al., 2014) had already been replicated qualitatively by BOLD imaging (Ulrich et al., 2016b). As functional challenge, we used again the flow paradigm with mental arithmetic tasks of different

* Corresponding author. Section Neuropsychology and Functional Imaging, Department of Psychiatry, University of Ulm, Leimgrubenweg 12-14 D-89075, Ulm, Germany.

E-mail address: martin.ulrich@uni-ulm.de (M. Ulrich).

<https://doi.org/10.1016/j.ynrp.2022.100139>

Received 8 June 2022; Received in revised form 5 October 2022; Accepted 5 October 2022

Available online 10 October 2022

2666-9560/© 2022 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

difficulty levels to operationalize the three conditions of boredom, flow, and overload. The computational realization permits online adaptation of individual difficulty levels to reach the best possible balance between actual task affordances and individual skills and/or abilities. Furthermore, solving math calculations comes with very clear goal settings and a clear feedback about whether the task could be solved or not. Therefore, the three necessary conditions for the emergence of flow are met in our flow paradigm (balance between the individual's skills and task difficulty; clear goals; and immediate feedback about performance; see Csikszentmihalyi et al., 2014, p. 232).

In order to address the replication character of the present study quantitatively, we followed recent recommendations by Ly et al. (2019) using the so-called replication Bayes factor (BF; see also Verhagen and Wagenmakers, 2014) to compute the evidence that the flow effect is present or absent in the replication attempt, taking the data of the original study fully into account. This strategy implies that the posterior probability from the original study is used as informed prior for the replication experiment, which is in accordance with Bayesian parameter updating. The resulting replication BF is then quantifying “the relative predictive adequacy of the null hypothesis versus an alternative hypothesis that is completely informed by the knowledge of the parameter obtained from the first study” (Ly et al., 2019, p. 2501).

According to the arrangement of the psychological conditions, that is, boredom, flow, and overload, two central neural flow effects were of interest. An inverted U-shaped effect, predicting greater neural activation during the flow condition relative to its boundary conditions, and a U-shaped flow effect where neural activation during the flow condition is decreased relative to the boundary conditions of boredom and overload. As in the reference study (Ulrich et al., 2016b), in the present investigation, we also employed additional measurements to indicate the emergence of flow, that is, experience ratings and electrodermal activity (EDA) for which the same logic of replication Bayes factors was applied as outlined above.

2. Material and methods

2.1. Participants of the replication study

Forty-one healthy, male students from the local university, with an average age of 22.9 years (standard deviation, SD: 2.6 years), participated in the study. Participants did not report any psychiatric/neurological disorders or contraindications with regard to MRI. The study was in accordance with the Declaration of Helsinki, and approved by the Ethics Committee at Ulm University. Written informed consent was obtained prior to the experiment. Financial compensation was 40 EUR.

2.2. Experimental task

Subjects performed visually presented mental arithmetic tasks as in our previous study (Ulrich et al., 2016b). On each trial, a math expression appeared in black font above an on-screen keyboard in the center of the screen. Stimuli were presented on white background on a 32 inches MR-compatible LCD screen (NordicNeuroLab AS, Bergen, Norway) at a resolution of 1024 × 768 pixels. The on-screen keyboard was controlled by an MRI-compatible trackball (NATa TECHNOLOGIES, Coquitlam, Canada). Participants were asked to sum the presented two or more numbers in their mind and to enter the result as accurately and as fast as possible using the on-screen keyboard. After a period of 10 s per each math calculation, or when participants had submitted the result, there was a break of 1 s during which the string “xxx + x” appeared on the screen. Calculations and intermittent breaks were presented until the block length of 30 s had elapsed. Depending on the specific experimental condition, task difficulty was either very low (boredom or “B” condition) or was challenging with dynamic adjustment of task difficulty to participants' individual skill level (flow or “F” condition) or was overwhelming due to very high task difficulty (overload or “O” condition). In

the boredom condition, subjects performed addition tasks involving always two numbers only, which were randomly drawn with the following restrictions: The first addend belonged to the interval of 100 to 109, the second addend to the interval 1–9, and the result was between 101 and 110. In the flow condition, task demands adapted automatically and continuously to participants' level of skill that had been estimated beforehand in a preparation phase (see below), and used as starting level. When the result was correct, task difficulty increased by one level, and when the result was incorrect, difficulty decreased by one level. Adjustment of task difficulty was achieved by two alternate mechanisms: In case the last addend had only one digit, the level increased by changing the last addend to a two-digit number in the next calculation. A further increase was accomplished by adding an additional one-digit number to the mathematical expression, etc. Analogously, a decrease in task difficulty was achieved by the two mechanisms outlined above but in reverse order. In the overload condition, the starting level of task difficulty was set three levels higher than the starting level of the flow condition. Levels were adjusted in the manner described above, but were not permitted to fall below the starting level. Our previous study (Ulrich et al., 2016b) had confirmed that task demands in the overload condition are sufficiently higher than participants' average level of skill, while minimizing the probability of permanent frustration. Subjects did not receive explicit feedback about response accuracy in any of the conditions. The experiment began with a block of a rest condition where subjects were asked to fixate a black cross, centered on a white screen while avoiding unnecessary movements. Afterwards, nine blocks per task condition, with intermitting rest blocks, were performed. Both task and rest blocks lasted 30 s. There were two predefined block sequences counterbalanced across participants: BFOBFBFOBFBFOBFBFOBFBFOBFB (rest blocks between each of the B, F, O conditions are not shown). In the preparation phase, prior to scanning, participants underwent two practice blocks. The first block consisted of an extended version of the B condition, lasting 3 min, to make participants familiar with the trackball. After a short break of 15 s, a modified version of the flow condition was performed, aimed at determining participants' individual level of skill: Starting at the lowest possible level, in the course of 4 min task demands were automatically and continuously adapted to participants' skills. The average task level pertaining to the last 25% of results was used as the starting level for the flow condition in the experiment. The computer algorithms generating and presenting the math calculations, analyzing the results and performing the level adjustments were programmed in Scala version 2.9.01 and run in Java Runtime Environment version 6.0.05 on a standard PC with 64 bit Windows 7 Service Pack 1.

2.3. MRI data acquisition

Magnetic resonance imaging data were acquired on a 3 T MAGNETOM Prisma (Siemens AG, Erlangen, Germany) with a 64 channel head/neck coil. During the experiment, an echo-planar pulse sequence (EPI) was applied to measure the T2*-weighted BOLD signal. The following parameters were used: repetition time (TR) = 2000 ms, echo time (TE) = 33 ms, flip angle = 90°, Parallel Acquisition Technique (PAT) factor = 2 (GRAPPA mode), field of view (FOV) = 220 mm, matrix size = 90 × 90, number of slices: 32, slice orientation: transversal, acquisition: ascending, slice thickness: 3.0 mm, interslice gap = 1.0 mm, bandwidth = 2136 Hz/pixel, voxel size = 2.44 mm × 2.44 mm × 4.00 mm. In the course of about 27.5 min, 825 EPI volumes were acquired. To obtain a high resolution 3D-T1-weighted structural image for later coregistration purposes, a magnetization prepared rapid acquisition gradient echo sequence was employed: TR = 2300 ms, TE = 2.61 ms, inversion time = 1100 ms, bandwidth = 290 Hz/pixel, flip angle = 9°, FOV = 256 mm, matrix size = 256 × 256, voxel volume = 1 mm³, slice orientation: sagittal; PAT factor = 2 (GRAPPA mode); scan time about 5 min.

2.4. Flow experience ratings

After the experiment, outside the MR scanner, participants rated questions assessing their subjective flow experiences during the fMRI experiment. Questions were presented in written form on a computer screen, and responses (mouse clicks) were automatically recorded by the computer program. As in our previous studies, presentation of the following three Likert-scaled statements was intended: "I would love to solve math calculations of that kind again", "Task demands were well matched to my ability" and "I was thrilled". Unfortunately though, due to a misunderstanding during the preparation of the study, the item "Task demands were well matched to my ability" had been erroneously replaced by the item "I had the necessary skill to solve the calculations successfully". Due to its inherent semantic ambiguity, this item did not show the same discriminatory power between the three conditions as the other two items. This is consistent with our very first study (Ulrich et al., 2014), where this item was also part of a questionnaire, and demonstrated insufficient between-conditions discrimination, too.

Consequently, as only two of the three items that typically form the subjective flow index had been implemented correctly, flow experience ratings were not further analyzed for the present purpose of replication (nevertheless, averaged data of the remaining items are visualized in Fig. S1).

2.5. Measurement of electrodermal activity

To obtain a marker of sympathetic activity in response to the task conditions, participants' EDA was measured during the fMRI experiment with the same setup as previously (Ulrich et al., 2016b) reported. Except for the electrode paste, hardware and software described below were from Brain Products GmbH (Gilching, Germany). Two reusable sintered silver/silver chloride electrodes (as part of the "GSR MR Set") were prepared with TD-246 Skin Conductance Electrode Paste (0.5% saline; Discount Disposables, St. Albans, VT, USA) and placed on the volar middle phalanges of the left index and middle fingers. The electrodes were connected to the "Galvanic Skin Response (GSR) sensor for fMRI" relying on the constant voltage (0.5 V) measuring principle. The signal was passed on to the "ExG AUX Box" before entering the amplifier "BrainAmp ExG-MR-16". Power was supplied by a rechargeable 5.6 V "PowerPack" battery. The amplified signal left the MR cabin through a fiber optic cable to undergo optoelectric conversion by the "USB 2 Adapter" connected to a standard laptop. The EDA signal was recorded at a sampling rate of 5000 Hz, and low-pass filtered (cutoff at 250 Hz) using the software BrainVision Recorder 1.20.0506 running under Windows XP Professional version 2002 Service Pack 3.

2.6. Data analysis

2.6.1. fMRI data - task-based activation

Due to advances of analysis pipelines during the past years, preprocessing and statistical analyses of fMRI data were performed with the same computational setup for both the previous data and the actual replication data. The most notable variation introduced was incorporation of the Computational Anatomy Toolbox 12 (CAT12.8.1; Gaser et al., in review) for normalization of T1 weighted images and the fMRI data of both studies. Otherwise, preprocessing and statistical analyses were performed under MATLAB 2021a (MathWorks Inc., Natick, MA, USA) with Statistical Parametric Mapping (SPM12 r6225, Wellcome Department of Cognitive Neurology, London, UK). Preprocessing started with segmentation of individual T1 images using CAT12 with default settings (except for voxel size of normalized images, set to 1 mm). This step produced bias, noise and global intensity corrected T1 images both in native and in normalized (Montreal Neurological Institute, MNI) space as well as forward deformation fields (native → normalized). Functional MRI data were first slice time corrected and spatially realigned. The resulting mean functional image was then co-registered to

the individual, CAT12-processed T1 image in native space and the co-registration matrix was applied to all other functional images. Afterwards, the functional MR images were MNI-normalized by applying the forward deformation fields from initial CAT12 segmentation as described above. Functional images were resampled to 2 mm × 2 mm × 2 mm, and smoothed using a Gaussian kernel with 8 mm full width at half maximum.

For each subject a General Linear Model (GLM) was set up by specifying the onsets and durations (30 s) of the task blocks for B, F and O as separate regressors. The spatial realignment parameters from each session were added to the design matrix as regressors of no interest. Resulting boxcar functions were convolved with the canonical hemodynamic response function. To remove low-frequency scanner drifts, data were high-pass filtered with a frequency cutoff at 128 s, and an autoregression model of polynomial order 1 was used to account for temporally correlated residual errors.

For reasons outlined below, two random effects group analyses were computed, one for the original study (Ulrich et al., 2016b; N = 23), and one for the combination of the previous and the actual study (N = 64). For both analyses, a flexible factorial design with Condition as within-subject factor was set up. A factor Subjects was added to the design matrix to account for interindividual variance of estimated neural activation during each task condition against baseline. For each statistical design, two one-sided t-contrasts were computed to test for between-condition differences in form of polynomial trends of second order, that is, mirroring an inverted U-shaped (invU) and a U-shaped (U) activation pattern (see Introduction).

2.6.2. fMRI data - replication analysis

To address the replication character of the present study, we made use of the replication Bayes factor as outlined in Ly et al. (2019). Usually, this replication BF can be computed by multiplying the BF from the first study with the BF of the replication study, whereby for calculation of the replication study's BF the posterior distribution from the first experiment is used as a prior distribution for the test of the replication experiment. Fortunately, as outlined by Ly et al. (2019), the process of parameter updating can be replaced by updating the evidence, which practically translates into the computation of two BF, the one for the previous study (original study BF), and the one for the combination of both studies, that is, the previous and the actual replication study (complete BF). By dividing both BF (complete BF/original study BF) the replication BF can be calculated (see Ly et al., 2019, pp. 2501–2502).

For calculation of the necessary BF for each combination of factors Study (original, complete) and Effect (invU, U), the $1/(-e * p * \ln(p))$ calibration, proposed by Sellke et al. (2001) (see also Bayarri et al., 2016; Held and Ott, 2018; Benjamin and Berger, 2019), was used, where the constant e denotes the base of the natural logarithm ($\ln$). This calibration depends directly on the p-values (p) associated with the t-contrasts for the invU and U-shaped flow effects, and reflects the highest possible BF consistent with the observed p-value (Benjamin and Berger, 2019), the so-called upper bound BF (BFB). To this end, unthresholded t-value maps from the corresponding t-contrasts were transformed into maps of p-values. Transformation was achieved using the "Transform SPM Maps" functionality as implemented in CAT12, and replacing the expression "t2x = 1-spm_Tcdf(Z,df(2))" in line 498 of the function "cat_stat_spm2x" by "t2x = tcdf(Z,df(2),'upper')" because the latter MATLAB built-in function gives higher precision for t-values corresponding to z-values above ≈ 8.21 . From the resulting maps p-values of $p \geq 1/e \approx 0.37$ were excluded, since the above calibration, $1/(-e * p * \ln(p))$, is defined only for p-values below this limit. In the next step, the calibration formula was applied and the resulting BFBs from the complete and original studies for each specific flow effect (invU, U) were divided (complete BFB/original study BFB) to compute the replication BF of each effect. Finally, the natural logarithms of the replication BF were computed and overlaid on a T1-weighted MR-image, computed as the average of stereotactically normalized individual T1-weighted

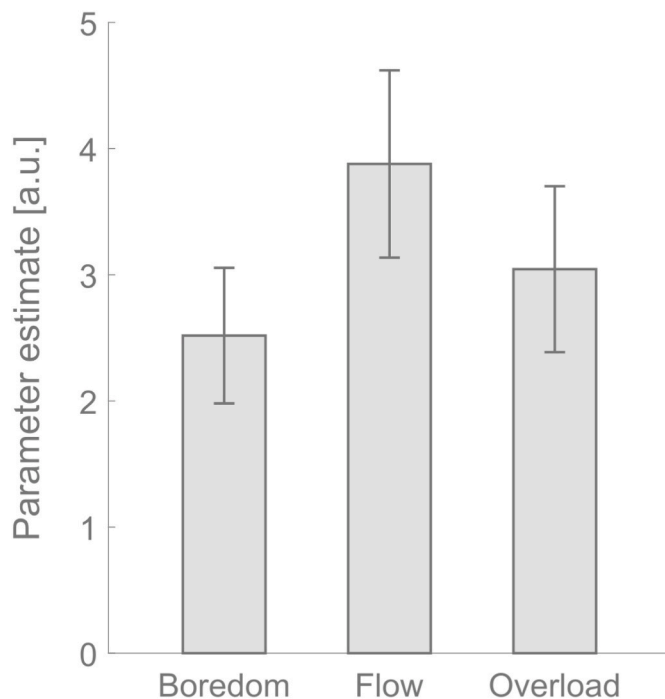


Fig. 1. Mean ($N = 63$) parameter estimates representing electrodermal activity (arbitrary unit, a.u.). Error bars indicate standard error of the mean.

3D-volumes from all participants in the previous and the actual study ($N = 64$). Overlaying was performed using Chris Rorden's MRICron (<https://www.nitrc.org/projects/mricron>; Rorden and Brett, 2000). Overlay images for the invU and U flow effects in the Results section are shown for natural logarithms of BF values ($\ln(\text{BF})$) above 5.704, corresponding to Bayes Factors >300 , which can be interpreted as decisive evidence for the effect, depending on previous categorizations of Bayes

factors (Held and Ott, 2018). For tabulation of brain regions with MNI coordinates of peak replication BF, the customized MATLAB function “get_cluster_maxima”, provided by Lotter (2021), was used.

2.6.3. EDA data

Raw data were converted into the Brain Vision Data Exchange Marker File format using BrainVision Analyzer 2.0.4.368 (Brain Products GmbH, Gilching, Germany). Using in-house written software under MATLAB R2009b, data were bandpass filtered with a third-order Butterworth filter and cutoff frequencies of 0.0078 Hz and 0.083 Hz, respectively, to remove slow and fast changes in skin conductance level. Data were then downsampled to a rate of 5 Hz, averaged in the interval between two consecutive TTL triggers to temporally align with the MR sampling rate (TR), and transformed into binary ANALYZE (7.5) file format to be further processed within the software framework of SPM8. Akin to the analysis of fMRI data, we used a temporally extended response function as predictor in a GLM approach. The underlying canonical skin conductance response function (SCRf) was set up as described by Bach et al. (2010), which is basically a modeled Gaussian smoothed exponential function with empirically derived estimates for event-to-peak time, rise time and two exponential constants approximately defining the decay. This canonical SCRf was convolved with typical boxcar functions with duration of 30 s. Individual first-level contrast estimates of the magnitude of skin conductance change against implicit baseline (fixation cross) were averaged for each condition and then linearly contrasted ($-B + 2F - O$) to test for the invU effect using a one-sample t -test. As this test revealed a significant effect above zero ($t(40) = 3.72$; $p < 0.001$), data from the actual replication study were combined with linearly contrasted EDA data from the original study to compute the complete BF as follows. First, two one-sample t -tests were computed, one for the invU EDA effect of the original study, and one for the complete study to obtain the p -values of each t -value for the effect of the linearly weighted contrast ($-B + 2F - O$, i.e., invU effect). Both p -values were transformed into the BFB by the formula $1/(-e * p * \ln(p))$, and then divided (complete BFB/original study BFB) to obtain the replication BF.

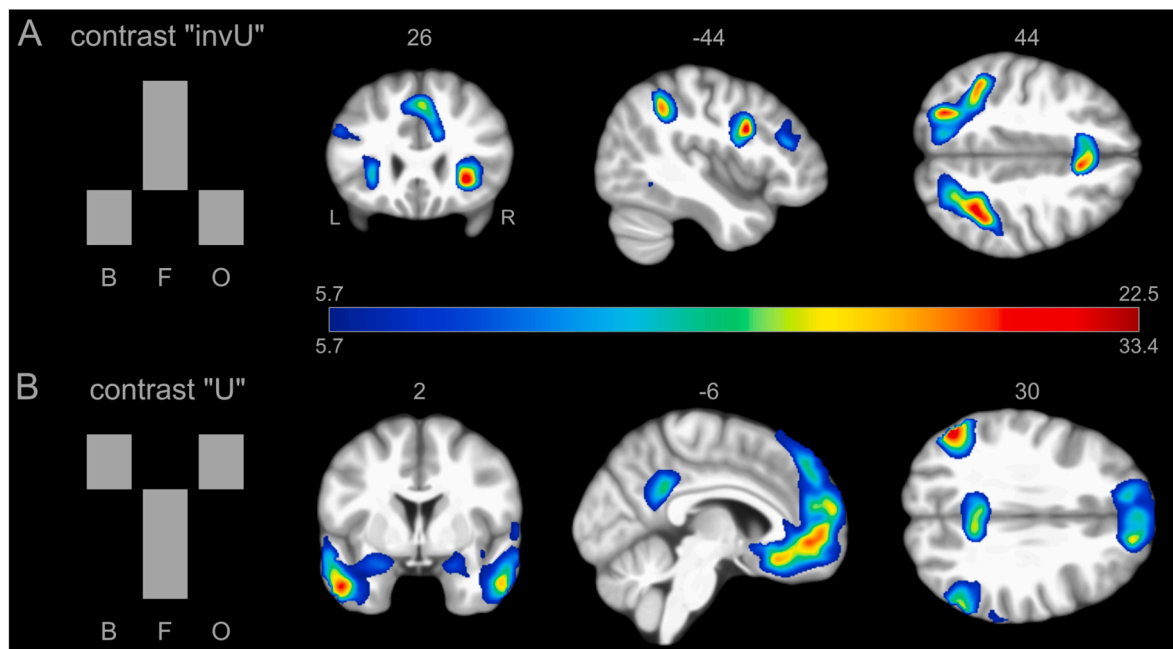


Fig. 2. Replication of flow-associated neural activation patterns. **A.** Inverted U-shape, reflecting greater brain activation associated with the flow condition (F) compared with boredom (B) and overload (O) conditions (contrast: $-B + 2F - O$). **B.** U-shape, revealed by the inverse contrast ($B - 2F + O$). The logarithmized replication Bayes factors are color-coded for the ranges 5.7–22.5 (invU effect) and 5.7–33.4 (U effect), respectively. The numbers above each slice refer to coordinates in MNI space. Effects were overlaid on participants' mean T1 image ($N = 64$) using MRICron (<https://www.nitrc.org/projects/mricron>; Rorden and Brett, 2000). L: left, R: right. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 1

Brain regions showing greater activation during the flow condition than during the boredom and overload conditions, revealed by the invU-shaped contrast [-B + 2F - O]. The replication Bayes factor map was thresholded such that $\ln(\text{BF}) > 5.704$.

Brain region	Cluster size (in voxels)	ln (BF)	Peak voxel (MNI space)		
			x	y	z
Cerebellum (R)	2613	22.55	6	-74	-26
Cerebellum (L)		19.31	-8	-74	-24
Cerebellum (L)		15.54	-28	-62	-30
Cerebellum (R)		13.89	26	-64	-28
Inferior parietal lobule (R)	2121	21.85	34	-50	42
Inferior parietal lobule (R)		20.81	42	-44	46
Inferior parietal lobule (R)		20.80	38	-48	44
Middle occipital gyrus (R)		17.57	30	-66	36
Anterior insula (R)	584	21.57	30	26	-4
Inferior parietal lobule (L)	2373	21.41	-28	-70	42
Inferior parietal lobule (L)		20.92	-36	-50	38
Precuneus (L)		16.72	-10	-72	52
Middle occipital gyrus (L)		13.91	-32	-80	32
Inferior frontal gyrus, opercular part (L)	1704	21.36	-44	6	28
Inferior frontal gyrus, opercular part (L)		16.50	-32	4	30
Middle frontal gyrus (L)		13.89	-24	10	58
Thalamus (L)		12.24	-16	-8	12
Middle cingulate cortex (R)	1606	20.59	6	18	44
Middle frontal gyrus (R)		18.51	28	8	56
Supplementary motor area		17.60	0	16	48
Anterior insula (L)	367	18.88	-32	22	2
Middle frontal gyrus (R)	765	17.95	48	38	24
Inferior frontal gyrus, triangular part (R)		17.85	46	38	28
Middle frontal gyrus (R)		15.30	38	36	18
Inferior temporal gyrus (L)	187	16.34	-52	-58	-12
Inferior frontal gyrus, triangular part (L)	376	12.97	-40	32	20
Inferior frontal gyrus, triangular part (L)		9.62	-50	28	28
Inferior frontal gyrus, opercular part (R)	264	12.72	42	10	28
Inferior temporal gyrus (R)	84	12.65	52	-52	-12
Middle orbitofrontal gyrus (R)	111	11.52	20	48	-20
Midbrain (L)	47	8.75	-8	-26	-8
Middle orbitofrontal gyrus (L)	50	8.41	-22	48	-16

3. Results

3.1. Electrodermal activity

Replication analysis of EDA data yielded a replication BF of 271 ($\ln(\text{BF}) = 5.60$). For illustration, the inverted U-shaped flow effect is depicted in Fig. 1 based on data of the complete study ($N = 63$ due to one obvious outlier as reported in the original study (Ulrich et al., 2016b)).

3.2. Replication of flow-associated invU-shaped neural activation pattern

Fig. 2A shows a map of replication BF for the inverted U-shaped flow activation pattern where replication BF were at least above 300 ($\ln(\text{BF}) = 5.704$) indicating decisive evidence for the replication of this effect. All brain regions with MNI coordinates of peak $\ln(\text{BF})$ are tabulated in Table 1.

3.3. Replication of flow-associated U-shaped activation pattern

Fig. 2B shows a map of replication BF for the U-shaped flow activation pattern where replication BF were at least above 300 ($\ln(\text{BF}) = 5.704$) indicating decisive evidence for the replication of this effect. All brain regions with MNI coordinates of peak $\ln(\text{BF})$ are tabulated in Table 2.

Table 2

Brain regions showing lower activation during the flow condition than during the boredom and overload conditions, revealed by the U-shaped contrast [-B + 2F + O]. The replication Bayes factor map was thresholded such that $\ln(\text{BF}) > 5.704$.

Brain region	Cluster size (in voxels)	ln (BF)	Peak voxel (MNI space)		
			x	y	z
Angular gyrus (L)	7384	33.39	-56	-66	32
Middle temporal gyrus (L)		32.24	-64	-16	-16
Inferior temporal gyrus (L)		30.38	-52	4	-34
Hippocampus (L)		20.57	-26	-14	-20
Middle temporal gyrus (L)		15.17	-58	-30	-6
Amygdala (L)		12.77	-30	0	-18
Putamen (L)		8.10	-22	10	-12
Medial orbitofrontal gyrus (L)	10362	28.51	-6	48	-4
Superior frontal gyrus (L)		26.61	-16	44	46
Rectal gyrus (L)		26.48	-6	28	-16
Superior frontal gyrus (R)		25.86	16	52	32
Medial superior frontal gyrus (L)		24.58	-10	58	20
Medial orbitofrontal gyrus		23.98	0	40	-10
Inferior temporal gyrus (R)	7310	28.47	54	-4	-28
Middle temporal gyrus (R)		25.90	62	-58	20
Middle temporal gyrus (R)		23.66	66	-16	-12
Hippocampus (R)		16.11	26	-14	-20
Putamen (R)		12.83	20	8	-8
Amygdala (R)		10.97	22	2	-20
Posterior cingulate cortex (R)	1194	23.17	8	-52	32
Posterior cingulate cortex (L)		21.77	-2	-48	30
Cerebellum (L)	62	14.52	-26	-80	-34
Cerebellum (R)	33	12.41	30	-82	-32
Middle cingulate cortex	162	11.06	0	-22	40
Middle cingulate cortex (L)		9.49	-2	-12	38
Posterior insula (L)	67	8.48	-38	-18	16
Inferior frontal gyrus, triangular part (R)	17	7.61	58	30	10
Inferior frontal gyrus, triangular part (L)	10	6.60	-58	26	6
Postcentral gyrus (R)	50	6.36	56	-8	40
Posterior insula (R)	11	6.31	38	-14	18
Postcentral gyrus (R)	11	6.30	48	-22	64

4. Discussion

The present study investigated the quantitative replication of the two major neural flow effects when employing our previously established functional paradigm to induce experimentally the emergence of flow. Following the relational operationalization of flow, this paradigm realizes three conditions, called boredom, flow and overload, and permits contrasting of neural activation of the flow condition against its boundary conditions. Two quadratic effects of neural activation have repeatedly been reported before: an inverted U-shaped relationship indicating greater neural activation for flow relative to boundary conditions, and its inverse, a U-shaped relationship.

For the purpose of replication, functional magnetic resonance imaging data from a fresh sample of 41 healthy male subjects were acquired. As in the original study used as reference (Ulrich et al., 2016b), read-out of neural activation was based on BOLD imaging. In order to quantify the replication effect, the replication Bayes factor as suggested by Verhagen and Wagenmakers (2014) was employed. Thankfully to later work by Ly et al. (2019), this replication Bayes factor can practically be implemented by comparing the Bayes factor for each effect (invU, U) from the original study against the Bayes factor of the same flow effect computed for the combined study. In the same way, quantitative replication analysis could also be performed for electrodermal activity as one of the two further read-outs previously used to test whether the experimental manipulation was effective. Unfortunately, quantitative replication analysis could not be performed for experiential ratings because, due to a simple misunderstanding, only two of the relevant items had been implemented for the replication study, thus

violating the idea of full experimental replication.

Replication of the electrodermal inverted U-shaped flow effect showed strong evidence for this effect, and can be interpreted that this measure is well suited to indicate robustness of the experimental manipulation and permitting interpretation of the functional imaging data.

For those data, replication Bayes Factors showed decisive evidence for the invU-shaped flow effect bilaterally in the anterior insula, dorsolateral prefrontal cortex, supplementary motor area, and posterior parietal cortex. The U-shaped flow effect was replicated with decisive evidence bilaterally in medial prefrontal cortex (including subgenual portions of the anterior cingulate cortex), middle and posterior cingulate cortex, ventral aspects of the striatum, amygdala, hippocampus, lateral temporal cortex, and angular gyrus.

As regards the functional interpretation of the brain regions involved we would like to draw the reader's attention to our previous work (Ulrich et al., 2016b). However, one more general aspect should be addressed in this context as a caveat when interpreting the inverted U-shaped effect: While the U-shaped activation pattern comes with almost no interpretational ambiguity, as reduced neural activation cannot be explained by the greater task demands that are usually necessary to induce flow, this is not so for the inverted-U effect. Here, the interpretation of how brain regions bearing this effect may relate to the emergence of flow is hindered because contributions of the experimental task per se (arithmetic calculation) cannot easily be disentangled. Nevertheless, some of those regions with an invU-shaped activation pattern are less likely to be particularly driven by mental arithmetic, e.g., the anterior insula.

While the present replication study indicated robustness of the two major flow effects on the neural level, and on the psychophysiological level, the strong limitation for this and our previous study is that all studies so far have been performed in men-only samples, so that generalizability to women awaits investigation. This restriction is due to the very first study where we investigated the neural correlates of experimentally induced flow for the very first time and where we wanted to preclude any hormonal influences on neural activation measures that may come from hormonal changes during the menstrual cycle (Dietrich et al., 2001; Fernández et al., 2003; Hausmann et al., 2002). Since then, this selection bias was propagated due to methodological reasons. It now imposes the strong necessity to set up a flow study with a women-only sample under strict hormonal control.

Another limitation is that estimation of Bayes factors for either study rely on a p-value based approximation via the formula $1/(-e * p * \ln(p))$. This yields an upper bound Bayes factor, showing “the highest possible BF consistent with the observed p-value” (Benjamin and Berger, 2019, p. 188). However, as reported by the same authors, while BFB is only an upper bound on the Bayes factor, Bayarri et al. (2016) had shown that, when BFB are calculated from real data, the BFB is often not that far from the BF, particularly when the logarithms of BFB are calculated. Furthermore, even if approximation of data-based BF by the p-value BFB may lead to an overestimation of BF, this does not affect calculation of the replication BF, since both BFB, the one from the original and the one from the complete study are set in relation to each other by division.

5. Conclusions

During the past years our previous results on the neural signature of experimentally induced flow have repeatedly been cited, which motivated us to investigate the robustness of those previous results by means of a replication study. Employing the concept of the replication Bayes factor (Ly et al., 2019; Verhagen and Wagenmakers, 2014), here we were in the position to show that both major flow effects on neural activation could be very well replicated. Of note, however, is that all studies so far have been conducted in men-only samples. Therefore, present results await empirical replication in at least two separate

women-only samples.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Author contributions

Martin Ulrich: Conceptualization, Methodology, Data curation, Formal analysis, Software, Visualization, Writing – original draft, review and editing. Filip Niemann: Conceptualization, Methodology, Investigation, Formal analysis, Writing – review and editing. Georg Grön: Conceptualization, Methodology, Formal analysis, Project administration, Supervision, Writing – original draft, review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ynirp.2022.100139>.

References

- Bach, D.R., Flandin, G., Friston, K.J., Dolan, R.J., 2010. Modelling event-related skin conductance responses. *Int. J. Psychophysiol.* 75 (3), 349–356. <https://doi.org/10.1016/j.ijpsycho.2010.01.005>.
- Bayarri, M.J., Benjamin, D.J., Berger, J.O., Sellke, T.M., 2016. Rejection odds and rejection ratios: a proposal for statistical practice in testing hypotheses. *J. Math. Psychol.* 72, 90–103. <https://doi.org/10.1016/j.jmp.2015.12.007>.
- Benjamin, D.J., Berger, J.O., 2019. Three recommendations for improving the use of p-values. *Am. Statistician* 73, 186–191. <https://doi.org/10.1080/00031305.2018.1543135>.
- Csikszentmihalyi, M., 1975. *Beyond Boredom and Anxiety: Experiencing Flow in Work and Play*. Jossey-Bass, San Francisco, CA.
- Csikszentmihalyi, M., 1990. *Flow: the Psychology of Optimal Experience*. Harper & Row, New York.
- Csikszentmihalyi, M., 2000. Happiness, flow, and economic equality. *Am. Psychol.* 55 (10), 1163–1164. <https://doi.org/10.1037/0003-066X.55.10.1163>.
- Csikszentmihalyi, M., Abuhamed, S., Nakamura, J., 2014. Chapter 15, flow. In: Csikszentmihalyi, M. (Ed.), *Flow and the Foundations of Positive Psychology*. Springer, Dordrecht, pp. 227–238.
- Dietrich, T., Krings, T., Neulen, J., Willmes, K., Erberich, S., Thron, A., Sturm, W., 2001. Effects of blood estrogen level on cortical activation patterns during cognitive activation as measured by functional MRI. *Neuroimage* 13 (3), 425–432. <https://doi.org/10.1006/nimg.2001.0703>.
- Fernández, G., Weis, S., Stoffel-Wagner, B., Tendolkar, I., Reuber, M., Beyenburg, S., Klaver, P., Fell, J., de Greiff, A., Ruhlmann, J., Reul, J., Elger, C.E., 2003. Menstrual cycle-dependent neural plasticity in the adult human brain is hormone, task, and region specific. *J. Neurosci.* 23 (9), 3790–3795. <https://doi.org/10.1523/JNEUROSCI.23-09-03790.2003>.
- Verhagen, J., Wagenmakers, E.J., 2014. Bayesian tests to quantify the result of a replication attempt. *J. Exp. Psychol. Gen.* 143 (4), 1457–1475. <https://doi.org/10.1037/a0036731>.
- Gaser, C., Dahnke, R., Kurth, K., Luders, E., Alzheimer's-Disease-Neuroimaging-Initiative. In Review. A Computational Anatomy Toolbox for the Analysis of Structural MRI Data. *Neuroimage*.
- Hausmann, M., Becker, C., Gather, U., Gunturkun, O., 2002. Functional cerebral asymmetries during the menstrual cycle: a cross-sectional and longitudinal analysis. *Neuropsychologia* 40 (7), 808–816. [https://doi.org/10.1016/s0028-3932\(01\)00179-8](https://doi.org/10.1016/s0028-3932(01)00179-8).
- Held, L., Ott, M., 2018. On p-values and Bayes factors. *Annu. Rev. Stat. Appl.* 5, 393–419. <https://doi.org/10.1146/annurev-statistics-031017-100307>.
- Lotter, Leon, 2021. Re: How to Extract Peak Coordinates from a Given Cluster with MATLAB as in the SPM Results Viewer? Retrieved on 17 May 2022 from: https://www.researchgate.net/post/How_to_extract_peak_coordinates_from_a_given_clus

- ter_with_MATLAB as in the_SPM_results_viewer/60221eebabf1b568b275758/citation/download.
- Ly, A., Etz, A., Marsman, M., Wagenmakers, E.J., 2019. Replication Bayes factors from evidence updating. *Behav. Res. Methods* 51 (6), 2498–2508. <https://doi.org/10.3758/s13428-018-1092-x>.
- Rorden, C., Brett, M., 2000. Stereotaxic display of brain lesions. *Behav. Neurol.* 12 (4), 191–200. <https://doi.org/10.1155/2000/421719>.
- Sellke, T., Bayarri, M.J., Berger, J.O., 2001. Calibration of p values for testing precise null hypotheses. *Am. Statistician* 55 (1), 62–71. <https://doi.org/10.1198/000313001300339950>.
- Ulrich, M., Keller, J., Grön, G., 2016a. Dorsal raphe nucleus down-regulates medial prefrontal cortex during experience of flow. *Front. Behav. Neurosci.* 10 (169) <https://doi.org/10.3389/fnbeh.2016.00169>.
- Ulrich, M., Keller, J., Grön, G., 2016b. Neural signatures of experimentally induced flow experiences identified in a typical fMRI block design with BOLD imaging. *Soc. Cognit. Affect Neurosci.* 11 (3), 496–507. <https://doi.org/10.1093/scan/nsv133>.
- Ulrich, M., Keller, J., Hoenig, K., Waller, C., Grön, G., 2014. Neural correlates of experimentally induced flow experiences. *Neuroimage* 86, 194–202. <https://doi.org/10.1016/j.neuroimage.2013.08.019>.
- Ulrich, M., Niemann, J., Boland, M., Kammer, T., Niemann, F., Grön, G., 2018. The neural correlates of flow experience explored with transcranial direct current stimulation. *Exp. Brain Res.* 236 (12), 3223–3237. <https://doi.org/10.1007/s00221-018-5378-0>.